



مدارس
للتنولوجيا
التطبيقية

IT Department
Web Development and Programming

تطوير المواقع والبرمجيات
Unit 20 PHP
الصف الثالث

Web Development and Programming
Question Bank

1st.

2025 - 2026

تطوير المواقع والبرمجيات

Unit 20 PHP

الصف الثالث

Unit	20	
Name	PHP Level 2	
<u>Goals / Outcomes</u>	By the end of this unit, students should be able to:: ➤ <u>Remembering</u> 1. Recall fundamental PHP concepts, including file handling, cookies, sessions, JSON, and filters. 2. Identify basic PHP functions such as fopen(), filter_var(), setcookie(), and CRUD-related functions. 3. Recognize the difference between validation and sanitization in PHP. ➤ <u>Understanding</u> 1. Explain how PHP integrates with web applications, handling files, cookies, sessions, and data exchanges through JSON. 2. Understand the role of PHP filters in data validation and sanitization. 3. Describe the process of working with databases using PDO and the CRUD model in PHP. ➤ <u>Applying</u> 1. Write PHP scripts to handle file uploads, manage cookies and sessions, and use filters for data validation. 2. Implement JSON for data exchange and work with databases to create, read, update, and delete records. 3. Utilize PHP error handling techniques and exception handling to build robust applications. ➤ <u>Analyzing</u> 1. Compare PHP file handling modes, session management techniques, and methods for handling user data. 2. Analyze how validation and sanitization impact data security in PHP. 3. Assess the structure and functionality of CRUD applications using PHP. ➤ <u>Evaluating</u> 1. Evaluate the efficiency and security of PHP scripts handling user input and data management. 2. Judge the effectiveness of PHP error handling and filtering mechanisms in improving code reliability. ➤ <u>Creating</u> 1. Develop complete PHP applications, integrating features such as file handling, cookies, sessions, JSON, and filters. 2. Build a functional CRUD application with user authentication and error handling.	
Knowledge	Code	Description

	TPK20	writing asynchronous code using various techniques
	TPK21	Basic routing and calling a controller method from a route
	TPK22	Analyze and solve common web applications tasks by writing PHP programs
Skill	Code	Description
	TPC5.1	Use database software tools to run queries and produce reports
	TPC5.2	Write PHP scripts to handle HTML forms

Unit Preface



المعرفة	الكود
writing asynchronous code using various techniques	TPK20

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	The time() function returns the current date formatted as 'Y-m-d'.	False
	The strtotime() function can parse English textual descriptions into a timestamp.	True
	In PHP OOP, an object is a template for a class.	False
	The \$this keyword is available outside of a class's methods.	False
	A constructor function in PHP is named construct().	False
	A destructor is automatically called when an object is destroyed or the script ends.	True
	A public property can be accessed from anywhere.	True
	A private property can be accessed by a class derived from the parent class.	False
	Encapsulation is the OOP principle for restricting access to internal details via access modifiers	True
	Inheritance allows a class to acquire properties and methods from another class.	True
	Polymorphism allows different objects to be used through the same interface.	True
	The getdate() function returns a Unix timestamp	False
	The checkdate() function is used to validate a Gregorian date. ()	True
	You must use the set_name() method to set a property if a constructor is defined.	False
	The date_default_timezone_set() function affects all date/time functions in the script.	True

Dear learner: Choose the correct answer.

No	Question	Answer
	Which function would you use to get the number of seconds since January 1, 1970? a) date() b) time() c) getdate() d) strtotime()	b
	What is the primary advantage of using a __construct function? a) To destroy the object b) To automatically initialize an object's properties upon creation c) To set the access modifier d) To make properties private	b
	Which access modifier restricts access to ONLY within the class itself? a) public b) final c) protected d) private	d
	The OOP principle that promotes code reuse by allowing a class to inherit from another is called: a) Encapsulation b) Inheritance c) Polymorphism d) Abstraction	b
	What keyword is used to create a new instance of a class (an object)? a) instance b) object c) new d) this	c
	Which function converts a string like "next Monday" into a Unix timestamp? a) date()	c

	b) time() c) strtotime() d) getdate()	
	Inside a class method, which keyword refers to the current object? a) self b) that c) this d) object	c
	Which principle restricts direct access to an object's data and uses methods to interact with it? a) Abstraction b) Inheritance c) Encapsulation d) Polymorphism	c
	How do you set the default timezone to "America/New_York"? a) set_timezone("America/New_York") b) date_default_timezone_set("America/New_York") c) default_timezone("America/New_York") d) timezone_set("America/New_York")	b
	What is a class in OOP? a) An instance of an object b) A template for objects c) A built-in PHP function d) A type of variable	b
	Which OOP principle allows objects of different classes to respond to the same method call in different ways? a) Encapsulation b) Inheritance c) Abstraction d) Polymorphism	d
	What will a <code>__destruct</code> function be called? a) When an object is created b) At the end of the script or when the object is destroyed	b

	c) Only when manually called d) At the beginning of the script	
	If a property is declared as protected, where can it be accessed? a) From anywhere b) Only within the class it is declared c) Within the class and by classes derived from it d) Only by the parent class	c
	The getdate() function returns its data in what format? a) A formatted string b) A Unix timestamp c) An associative array d) A serialized object	c

Match the PHP function on the left with its correct description on the right.

No	Function	No.	Description	Answer
1	date()	A	Returns an associative array with date/time information.	1 → c
2	time()	B	Parses an English textual datetime into a timestamp.	2 → e
3	strtotime()	C	Formats a local date and time into a specified string.	3 → b
4	getdate()	D	Validates a Gregorian date (e.g., checks for leap year).	4 → a
5	checkdate()	E	Returns the current Unix timestamp.	5 → d
6	__construct()	F	A special method called when a new object is created.	6 → f

Give the Scientific Name Questions:

No	Question	Answer
	A function that is automatically called when you create an object from a class.	constructor
	The keyword used to refer to the current object inside a class method.	\$this

	The access modifier that allows a property to be accessed from anywhere in the code.	public
	The OOP principle that helps keep code DRY by allowing classes to acquire properties from other classes.	inheritance
	The programming paradigm that is about creating objects containing both data and functions.	OOP
	A function that is automatically called at the end of a script or when an object is destroyed.	destructor
	The principle of restricting direct access to some of an object's components.	encapsulation
	An instance of a class.	object
	A template for objects, defining properties and methods.	class
	The principle that allows objects of different types to be treated as objects of a common supertype.	polymorphism

Complete the Sentence Questions:

No	Question	Answer
	The <code>date()</code> function is used to format a local and time.	date
	The _____ function returns the current Unix timestamp.	time()
	To create a new object, you use the _____ keyword.	new
	The _____ keyword refers to the current object inside a class method.	\$this
	A constructor function starts with two underscores: _____.	__construct
	A _____ property can only be accessed within the class itself.	private
	The OOP principle of _____ focuses on hiding complex details and showing only essential features.	abstraction
	The _____ function validates if a Gregorian date is correct.	checkdate()

	You can use <code>date_default_timezone_set()</code> to set the default timezone for all date/time functions in a script.	<code>date_default_timezone_set()</code>
	In OOP, a class is a <code>template</code> for objects.	<code>template</code>

Dear learner: Answer the following questions.

	Question	What is the difference between a class and an object?
	Answer	A class is a template or blueprint. An object is an instance of a class.

	Question	What does the acronym DRY mean in the context of OOP advantages?
	Answer	Don't Repeat Yourself. It means code is reusable and not duplicated.

	Question	Name the three access modifiers in PHP OOP.
	Answer	public, protected, private.

	Question	What does the \$this keyword represent inside a class method?
	Answer	It represents the current object instance calling the method.

المعرفة	الكود
Basic routing and calling a controller method from a route	TPK21

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	File handling in PHP can only be used to read files, not write to them.	False
	The enctype attribute for a file upload form must be set to multipart/form-data.	True
	The \$_FILES superglobal array is used to access information about an uploaded file.	True
	The move_uploaded_file() function is used to delete a file from the server.	False
	PDO stands for PHP Data Orientation.	False
	PDO can only be used to connect to MySQL databases.	False
	Prepared statements in PDO help prevent SQL injection attacks.	True
	The default error mode for PDO is PDO::ERRMODE_EXCEPTION.	False
	Cookies are stored on the server.	False
	The setcookie() function must be called before any HTML output is sent to the browser.	True
	The \$_COOKIE superglobal array is used to access cookie values.	True
	To delete a cookie, you set its expiration time to a past date.	True
	Sessions are stored on the client's browser.	False
	The session_start() function is used to begin a new session or resume an existing one.	True
	Session variables are stored in the \$_SESSION superglobal array.	True
	The session_destroy() function alone is sufficient to immediately unset all session variables.	False
	Cookies are more secure than sessions for storing sensitive data because they are stored on the server.	False

	The <code>session_unset()</code> function removes all session variables.	True
	The <code>isset()</code> function can be used to check if a cookie has been set.	True
	Sessions use a cookie on the client-side to store the session ID.	True

Dear learner: Choose the correct answer.

No	Question	Answer
	Which function is crucial for moving an uploaded file to a permanent location? a) <code>copy_file()</code> b) <code>move_uploaded_file()</code> c) <code>save_upload()</code> d) <code>store_file()</code>	b
	Which HTML form attribute is essential for enabling file uploads? a) <code>method="GET"</code> b) <code>action="upload.php"</code> c) <code>enctype="multipart/form-data"</code> d) <code>type="file"</code>	c
	Where is information about an uploaded file temporarily stored before being processed by PHP? a) <code>\$_FILES["file"]["tmp_name"]</code> b) <code>\$_FILES["file"]["perm_name"]</code> c) <code>\$_POST["file"]</code> d) <code>\$ GET["file"]</code>	a
	What is the primary advantage of using PDO over the older mysql extension? a) It's faster b) It provides a consistent interface for multiple databases c) It doesn't require a password d) It only works with MySQL	b
	Which PDO error mode is recommended for using exceptions for error handling?	c

	a) PDO::ERRMODE_SILENT b) PDO::ERRMODE_WARNING c) PDO::ERRMODE_EXCEPTION d) PDO::ERRMODE_STRICT	
	Which method is used to execute a prepared statement in PDO? a) run() b) execute() c) query() d) fetch()	b
	What is the main security benefit of using prepared statements? a) They make queries run faster b) They prevent SQL injection attacks c) They compress data d) They encrypt database connections	b
	Which function is used to create a cookie in PHP? a) create_cookie() b) make_cookie() c) setcookie() d) set_cookie()	c
	Which superglobal array is used to access a cookie's value? a) \$_GET b) \$_POST c) \$_REQUEST d) \$_COOKIE	d
	To delete a cookie, you set its expiration time to: a) time() + 3600 b) 0 c) A time in the past (e.g., time() - 3600) d) null	c
	Where is session data stored by default? a) On the client's browser b) In a database	d

	c) In a cookie d) On the server	
	Which function must always be called before using \$_SESSION variables? a) session_begin() b) start_session() c) session_start() d) session_init()	c
	Which superglobal array is used to store and access session variables? a) \$_COOKIE b) \$_SESSION c) \$_SERVER d) \$_STORAGE	b
	To remove all session variables but keep the session intact, you would use: a) session_destroy() b) session_remove() c) session_unset() d) unset(\$_SESSION)	c
	To completely destroy a session, including the session ID, you should use: a) session_unset() followed by session_destroy() b) Only session_destroy() c) unset(\$_SESSION) d) delete_session()	a
	What is stored on the client's browser to identify a user's session? a) All their session variables b) A session ID c) Their username d) A unique token	b
	Which of these is a key difference between sessions and cookies? a) Sessions are stored on the client, cookies on the server.	b

	b) Cookies are stored on the client, sessions on the server. c) Sessions are always persistent. d) Cookies cannot be deleted.	
	Which function checks if a variable is set and is not NULL? a) is_null() b) check() c) isset() d) empty()	c
	In a PDO connection string (DSN), what does mysql:host=localhost;dbname=testdb specify? a) The username and password b) The driver, host, and database name c) The error mode d) The prepared statement	b
	Which PDO method is best for executing a SELECT statement with variables? a) exec() b) query() c) prepare() and execute() d) run()	c

Give the Scientific Name Questions:

No	Question	Answer
	The PHP superglobal array that contains information about files uploaded via an HTTP POST method.	\$_FILES
	The function used to relocate an uploaded file from its temporary directory to a permanent location on the server.	move_uploaded_file
	A database access layer in PHP that provides a uniform interface for accessing multiple databases.	PDO

	A database feature supported by PDO that separates SQL logic from data, greatly enhancing security.	Prepared Statements
	The PDO attribute that must be set to PDO::ERRMODE_EXCEPTION to handle errors using exceptions.	PDO::ATTR_ERRMODE
	Small files of information that are stored on the client's browser to persist data between requests.	Cookies
	The PHP function used to create a new cookie or modify an existing one.	setcookie
	The superglobal array used to retrieve the value of a cookie that was sent from the client's browser.	\$_COOKIE
	A way to store temporary user information on the server, making it available across multiple pages.	Session
	The function that must be called at the beginning of every script that wishes to use or create session variables.	session_start
	The superglobal array specifically designed for storing and accessing session variables.	\$_SESSION
	The function used to completely remove all session data and destroy the session.	session_destroy

Complete the Sentence Questions:

No	Question	Answer
	The form attribute enctype must be set to _____ for file uploads to work.	multipart/form-data
	The _____ function is used to check if a file already exists on the server.	file_exists
	PDO is an abbreviation for _____.	PHP Data Objects

	To connect to a database with PDO, you need to create a new PDO instance with a connection string known as a _____.	DSN (Data Source Name)
	The _____ method of a PDO object is used to create a prepared statement. Answer:	prepare
	To set a cookie to expire in one day, you would set the expire parameter to time() + _____ seconds.	86400
	You should use the _____ function to check if a cookie is set before trying to use its value.	isset
	Session variables are stored on the _____, while cookies are stored on the _____.	server, client
	The _____ function removes all session variables but does not destroy the session itself.	session_unset
	The server identifies a user's session on subsequent requests using a unique _____ that is typically stored in a cookie.	session ID
	In the PDO DSN for MySQL, you specify the database driver using _____:.	mysql
	The _____ method of a PDO statement object is used to fetch a single row from a result set.	fetch

Match the PHP Superglobal with its purpose.

No	Superglobal	No	purpose	Answer
1	\$_FILES	A	Stores variables passed to the script via HTTP POST.	1→B
2	\$ SESSION	B	Stores information about uploaded files.	2→D
3	\$_COOKIE	C	Stores variables passed to the script via HTTP GET.	3→E

4	\$_POST	D	Stores session variables for the current user.	4→A
5	\$_GET	E	Stores variables passed to the script via HTTP cookies.	5→C
6	\$_SERVER	F	Stores information about headers, paths, and script locations.	6→F

Match the PDO-related term with its description.

No	term	No	description	Answer
1	DSN	A	The method used to send parameters to a prepared statement.	1→C
2	Prepared Statement	B	A value passed to setAttribute() to control how errors are handled.	2→D
3	execute()	C	A connection string that defines the database driver and details.	3→A
4	PDO::ATTR_ERRMODE	D	A pre-compiled SQL query that separates logic from data.	4→B
5	fetch()	E	The method used to retrieve a result row from a query.	5→E
6	PDOException	F	The type of exception thrown by PDO on errors.	6→F

Match the function with its action.

No	function	No	action	Answer
1	setcookie()	A	Starts a new or resumes an existing session.	1→C
2	session_start()	B	Moves an uploaded file to a new location.	2→A
3	move_uploaded_file()	C	Creates or modifies a cookie.	3→B

4	session_destroy()	D	Removes all session variables.	4→E
5	session_unset()	E	Completely destroys all data registered to a session.	5→D
6	isset()	F	Checks if a variable is declared and is not null.	6→F

Dear learner: Answer the following questions.

Question	What are the two essential attributes an HTML form must have to allow file uploads?
Answer	method="POST" and enctype="multipart/form-data".

Question	What is the purpose of the session ID?
Answer	It is a unique identifier stored on the client.

المعرفة	الكود
Analyze and solve common web applications tasks by writing PHP programs	TPK22

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement.

No	Question	Answer
	JSON is primarily used for styling web pages.	False
	The json_encode() function converts a JSON string into a PHP variable.	False
	A Try-Catch block is used for handling errors gracefully.	True
	In the example, divide(10, 0) will execute without causing an error.	False
	PHP filters are only used for validating data, not for sanitizing it.	False
	FILTER_VALIDATE_EMAIL checks if an email address is properly formatted.	True

	<code>FILTER_SANITIZE_NUMBER_INT</code> will remove all alphabetic characters from a string, leaving only numbers and digits.	True
	The filter <code>input()</code> function is used to filter data from a file.	False
	<code>FILTER_SANITIZE_SPECIAL_CHARS</code> converts special characters like <code><</code> and <code>></code> into HTML entities.	True
	The filter <code>var_array()</code> function is used to filter a single variable with multiple filters.	False
	You can set options like 'min_range' and 'max_range' with <code>FILTER_VALIDATE_INT</code> .	True
	In a Try-Catch block, the code in the catch section runs only if an error occurs in the try section.	True

Dear learner: Choose the correct answer.

No	Question	Answer
	What does JSON stand for? a) Java Standard Object Notation b) JavaScript Object Notation c) JavaScript Operational Network d) Java Oriented Numbering	b
	Which PHP function converts a PHP array or object into a JSON string? a) <code>json_convert()</code> b) <code>json_string()</code> c) <code>json_decode()</code> d) <code>json_encode()</code>	d
	What is the primary purpose of a Try-Catch block? a) To make code run faster b) To hide all errors from the user c) To handle potential errors without crashing the script d) To terminate the script safely	c
	In the context of PHP filters, what does "sanitization" mean? a) Checking if data meets a specific format b) Encrypting data for security	c

	c) Removing or encoding unwanted characters d) Deleting the input data	
	Which filter would you use to check if a value is a whole number? a) FILTER_SANITIZE_NUMBER_INT b) FILTER_VALIDATE_INT c) FILTER_VALIDATE_FLOAT d) FILTER_SANITIZE_SPECIAL_CHARS	b
	You have a string "123abc456". Using FILTER_SANITIZE_NUMBER_INT, what is the result? a) 123 b) 456 c) 123456 d) "abc"	c
	Which function is best for retrieving and filtering data from a POST form submission? a) filter_var() b) filter_input() c) filter_SANITIZE() d) get_post_input()	b
	What does the second parameter (true) in json_decode(\$jsonFile, true) do? a) Makes the function throw an error on failure b) Returns the data as an associative array instead of an object c) Validates the JSON before decoding it d) Encodes the JSON with pretty print	b
	Which part of a Try-Catch block contains the code that might generate an error? a) The catch block b) The finally block c) The error block d) The try block	d
	To validate multiple input variables at once, which function should you use? a) filter_var()	c

	b) filter_input() c) filter_var_array() d) array_filter()	
	Which PDOException is caught in the database connection example? a) A general Exception b) A file reading error c) A database connection error d) A division by zero error	c

Give the Scientific Name Questions:

No	Question	Answer
	A lightweight format for storing and exchanging data.	JSON
	The PHP function that converts a JSON string into a PHP variable.	json_decode
	A code block that acts as a safety net for handling runtime errors.	Try-Catch
	The type of PHP filter that checks if data meets specific criteria, like a valid email format.	Validation filter
	The function used to get a single external variable (e.g., from POST) and filter it.	filter_input
	The specific exception type caught when a database connection fails using PDO.	PDOException
	The term for the process of cleaning or filtering input data to make it safe.	Sanitization

Complete the Sentence Questions:

No	Question	Answer
	The json_encode() function converts a PHP _____ or array into a JSON string.	object
	To read the contents of a file into a string, you use the _____ function.	file_get_contents
	In a Try-Catch block, the code to handle an error is placed inside the _____ block.	catch

	The PHP function <code>filter_var()</code> is used to filter a variable.	single
	The <code>filter_input()</code> function's first parameter specifies the input source, such as <code>INPUT_POST</code> or <code>INPUT_GET</code> .	INPUT_POST
	When using <code>FILTER_VALIDATE_INT</code> , you can use the _____ option to set a minimum value.	min_range

167 Match the PHP Function to its Purpose.

No	Function	No.	Purpose	Answer
1	<code>json_encode()</code>	A	Converts a JSON string into a PHP variable	1→D
2	<code>json_decode()</code>	B	Gets and filters an external variable (e.g., from POST)	2→A
3	<code>filter_var()</code>	C	Reads the entire contents of a file into a string	3→E
4	<code>filter_input()</code>	D	Converts a PHP variable into a JSON string	4→B
5	<code>filter_var_array()</code>	E	Filters a single variable	5→F
6	<code>file_get_contents()</code>	F	Filters multiple variables at once	6→C

168 Match the Filter to its Primary Action.

No	Column (A)	No.	Column (B)	Answer
1	<code>FILTER_VALIDATE_EMAIL</code>	A	Encodes special characters into HTML entities	1→D
2	<code>FILTER_VALIDATE_URL</code>	B	Removes illegal characters from an email string	2→E
3	<code>FILTER_SANITIZE_EMAIL</code>	C	Checks if a value is a whole number	3→B
4	<code>FILTER_SANITIZE_NUMBER_INT</code>	D	Checks for a valid email format	4→F
5	<code>FILTER_SANITIZE_SPECIAL_CHARS</code>	E	Checks for a valid web address	5→A

6	<code>FILTER_VALIDATE_INT</code>	F	Removes all characters except digits and +/- signs	6→C
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Dear learner: Answer the following questions.

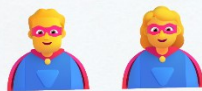
	Question	What are the two main purposes of PHP filters?
	Answer	To validate and to sanitize external input.

	Question	What will be the output of echo <code>json_encode(array("John", "Mariam"));</code>?
	Answer	<code>["John","Mariam"]</code>

NOTICE



If you come across a question that's been answered wrong, don't panic! Instead, be the hero we need and report it through this magical link.



IT department



<https://forms.gle/FJPz9tCnNtmoBJwH9>